Form Validation

- Form validation is the process of checking the data entered by a user before the form is submitted.

- Validation helps identify incorrect or incomplete input.
- It can provide a message to the user when the entered data is invalid.
- Validation can be performed before sending form data to the server.

Basic Form Validations

- Name
- Password
- Mobile
- Email
- Date

1. Name Validation

- Name validation checks whether the user has entered a name and whether the name contains acceptable characters.

- Name should not be empty and can be checked using a regular expression that allows alphabetic characters and spaces.

```
let name = document.getElementById("name").value;

let namePattern = /^[A-Za-z ]+$/;

if (name === "") {
   console.log("Name is required");
} else if (!namePattern.test(name)) {
   console.log("Enter a valid name");
} else {
   console.log("Valid name");
}
```

Name Validation Rules

- Name should not be empty.
- Name should contain alphabetic characters and spaces.
- Special characters and numbers can be rejected when the application requires a simple alphabetic name.

2. Password Validation

- Password validation checks whether the user has entered a password and whether it satisfies the required password rules.

- A common basic rule is to require a minimum number of characters. Applications can define stronger rules depending on their security requirements.

```
let password = document.getElementById("password").value;


if (password === "") {
   console.log("Password is required");
} else if (password.length < 8) {
   console.log("Password must contain at least 8 characters");
} else {
   console.log("Valid password");
}
```

Password Validation Rules

- Password should not be empty.
- A minimum length should be defined by the application.
- For stronger validation, applications may require uppercase letters, lowercase letters, numbers, and special characters.
- Passwords should not be displayed openly on the page.

3. Mobile Validation

- Mobile validation checks whether the entered mobile number follows the format expected by the application.

- For a simple example, a 10-digit numeric mobile number can be checked using a regular expression.

```
let mobile = document.getElementById("mobile").value;

let mobilePattern = /^[0-9]{10}$/;

if (mobile === "") {
   console.log("Mobile number is required");
} else if (!mobilePattern.test(mobile)) {
   console.log("Enter a valid 10-digit mobile number");
} else {
   console.log("Valid mobile number");
}
```

Mobile Validation Rules

- Mobile number should not be empty.
- For this basic example, exactly 10 digits are required.
- Only numeric digits are accepted.
- The exact mobile-number format should match the application's requirements and country rules.

4. Email Validation

- Email validation checks whether the entered value follows a basic email address pattern.

```
let email = document.getElementById("email").value;

let emailPattern = /^[^\s@]+@[^\s@]+\.[^\s@]+$/;
```

```
if (email === "") {
   console.log("Email is required");
} else if (!emailPattern.test(email)) {
   console.log("Enter a valid email address");
} else {
   console.log("Valid email");
}
```

Email Validation Rules

- Email should not be empty.
- A basic email pattern should be checked.
- The value should contain an appropriate local part, @ symbol, and domain part.
- Client-side pattern checking is only a basic format check; it does not prove that an email address exists.

5. Date Validation

- Date of Birth validation checks whether a date has been entered and whether it satisfies the application's rules.

- A simple validation can check that the field is not empty. A common additional rule is that the Date of Birth should not be a future date.

```
let dob = document.getElementById("dob").value;
let today = new Date();
let selectedDate = new Date(dob);


if (dob === "") {
   console.log("Date of Birth is required");
} else if (selectedDate > today) {
   console.log("Date of Birth cannot be a future date");
} else {
```

```
    console.log("Valid Date of Birth");
}
```

Date Validation Rules

- Date of Birth should not be empty.
- The selected date should be a valid date.
- The Date of Birth should not be in the future.
- An application may also define an age range when required.

Note:

Regular Expression (Regex)

A regular expression is a pattern used to match and validate text. In form validation, regular expressions are commonly used for fields such as name, mobile number, and email.

```
let pattern = /^[0-9]{10}$/;


console.log(pattern.test("9876543210"));
```

The test() method checks whether the supplied value matches the pattern.